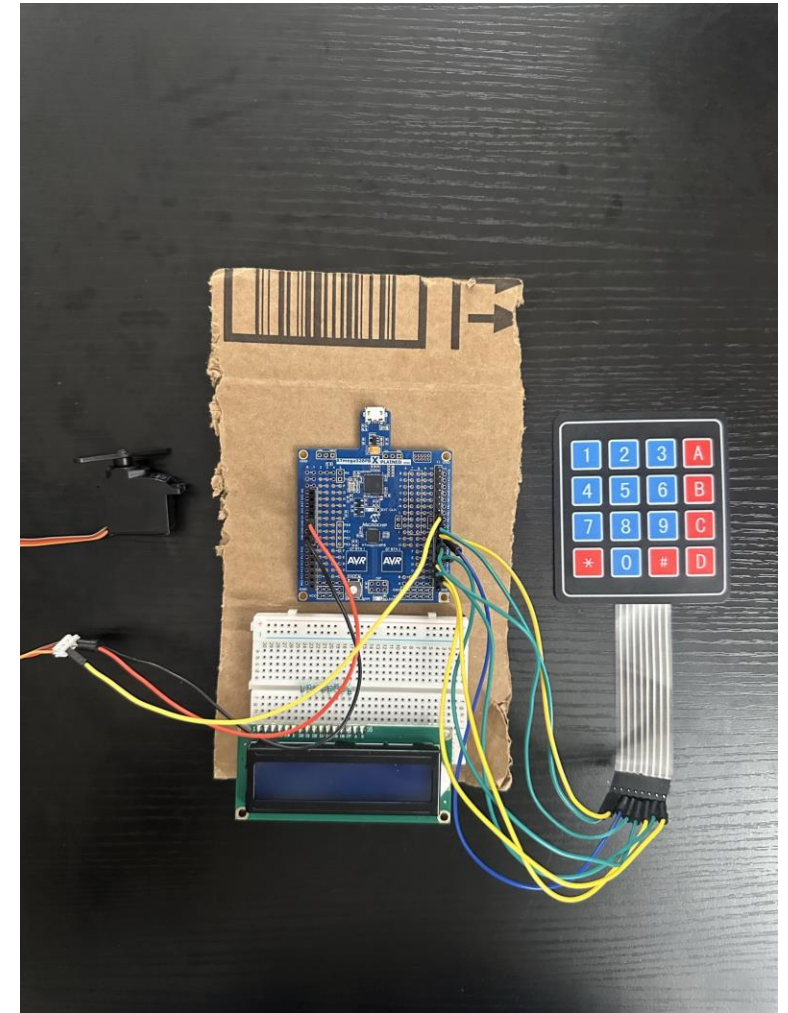

2 FACTOR AUTHENTICATION LOCK

Ethan Suttor, Dante Stewart, and Gavin Marvelle

FINAL PRODUCT

- 2 Factor lock; Keypad and Password in terminal
- Can be implemented in a wide variety of conditions



BOUNDARY CONDITIONS

- Small and versatile
- 5V power supply
- Must make use of interrupts and Interrupt Service Routine (ISR)
- Short turn around

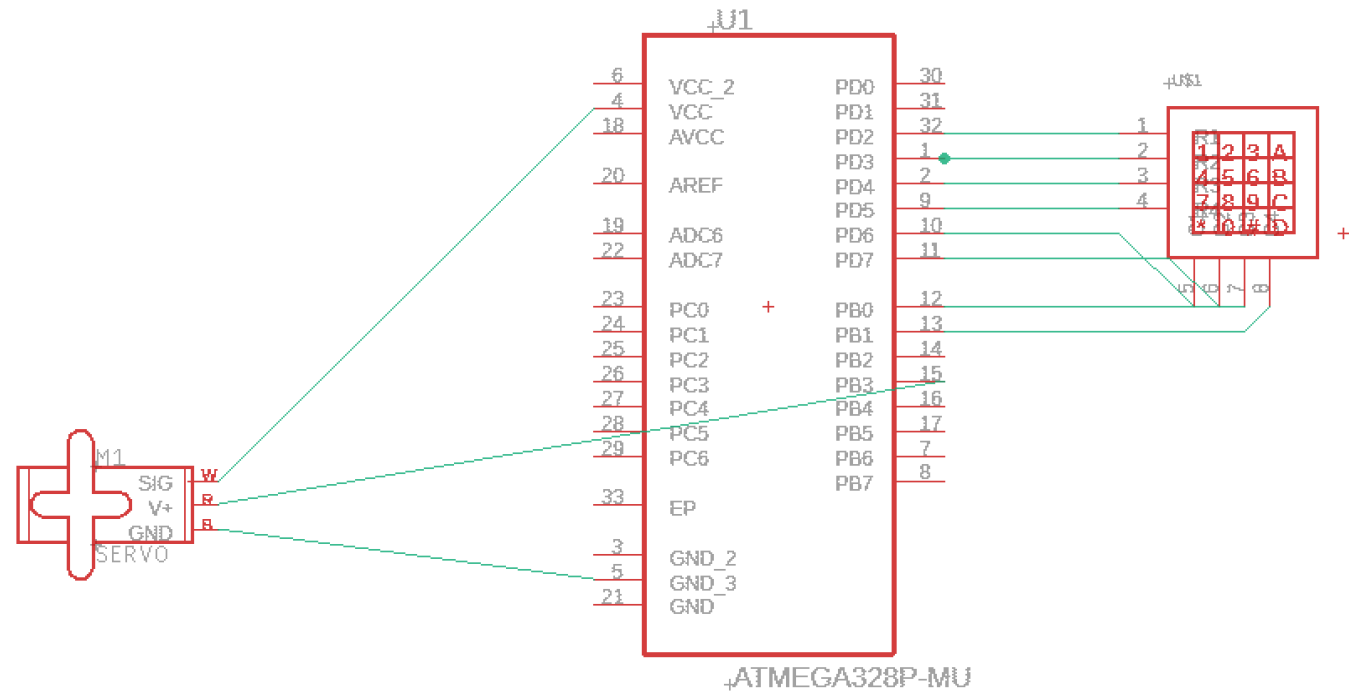
USE CONDITIONS



- Used when security is the top priority
- Numbers are too simple when short and can be easy to forget when long
- Keycards and key fobs can be stolen or lost

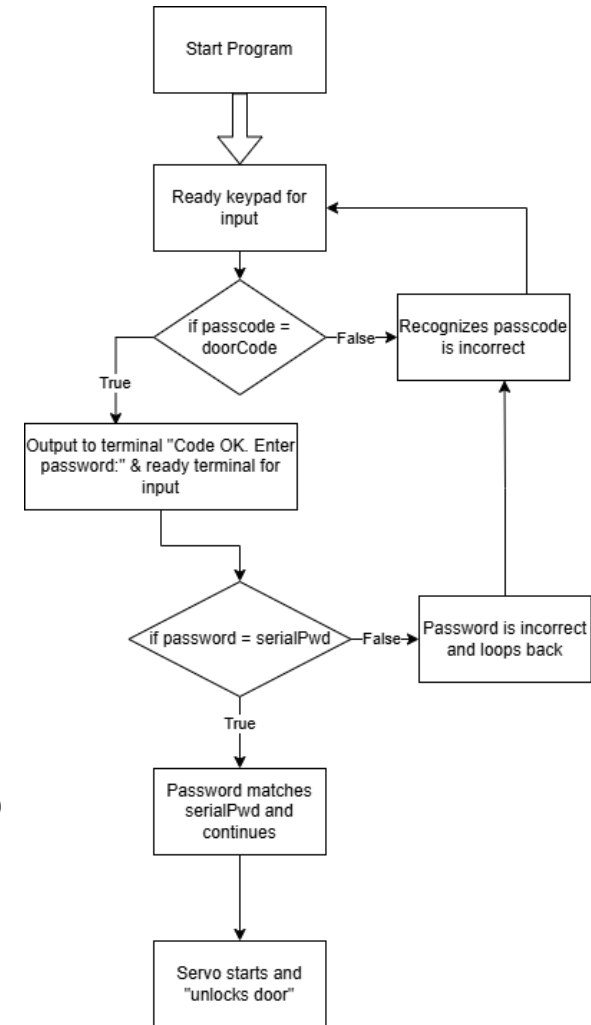
HARDWARE

- Membrane switch keypad
- Datan B1122 servo motor
- ATmega328PB Xplained board



SOFTWARE

- Debounce on the keypad
- 4-digit code match, prompt for terminal password over UART
- Verify serial password, then send 2.0 ms pulse to servo to unlock for 5 seconds



DEBOUNCE

```
// Debounce
uint8_t getKeyStable(char *out) {
    char a, b;
    if (!scan_keypad(&a)) return 0;
    _delay_ms(30);
    if (!scan_keypad(&b)) return 0;
    if (a == b) {
        *out = a;
        return 1;
    }
    return 0;
}
```

- 30ms wait time for each keypress
- Eliminated accidental keypresses

• 4-DIGIT CODE MATCH, TERMINAL PASSWORD OVER UART

- String compare & branching
- Uses strcmp(entry, doorCode) to detect a correct 4-digit entry, otherwise prints an immediate retry message.
- UART prompt for 2nd factor
- On match, calls uart_putstr("Code OK. Enter password:\r\n"); to prompt the user over serial before invoking the password-check routine.

```
while (1) {  
    if (getKeyStable(&key)) {  
        if (key != lastKey) {  
            lastKey = key;  
            uart_putchar(key);  
            entry[idx++] = key;  
            if (idx == strlen(doorCode)) {  
                uart_putstr("\r\n");  
                if (!strcmp(entry, doorCode)) {  
                    uart_putstr("Code OK. Enter password:\r\n");  
                    if (check_serial_pwd()) start_unlock();  
                    else uart_putstr("Wrong password. Try code again:\r\n");  
                } else {  
                    uart_putstr("Wrong code. Try again:\r\n");  
                }  
                idx = 0;  
                memset(entry, 0, sizeof(entry));  
            }  
        }  
    }  
}
```

OVERALL PROGRESS

- The overall this project was a success as it complete our goal of creating a lock with a password and key code.